National Institute of Technology Hamirpur
Department of Computer Science and Engineering
CS-414 Information Security and Privacy Lab
Assignment-1

Faculty: Dr. Ajay Kumar Mallick

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Session: Odd Sem, 2026-27

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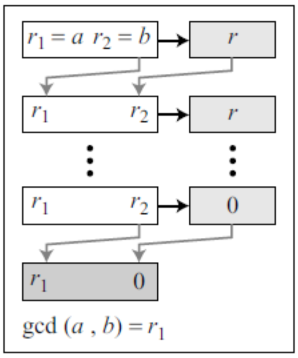
1. **Write a program to find the Greatest Common Divisor (GCD) of two integers using the Euclidean. Example, say GCD of 25 and 60 is 5. Print the GCD value.**

Algorithm.

1. Euclidean Algorithm:



a. Process

```
r1 ← a; r2 ← b;   (Initialization)
while (r2 > 0)
  {
   q ← r1 / r2;
    r ← r1 − q × r2;
    r1 ← r2; r2 ← r;
  }

   gcd (a, b) ← r1
```

b. Algorithm

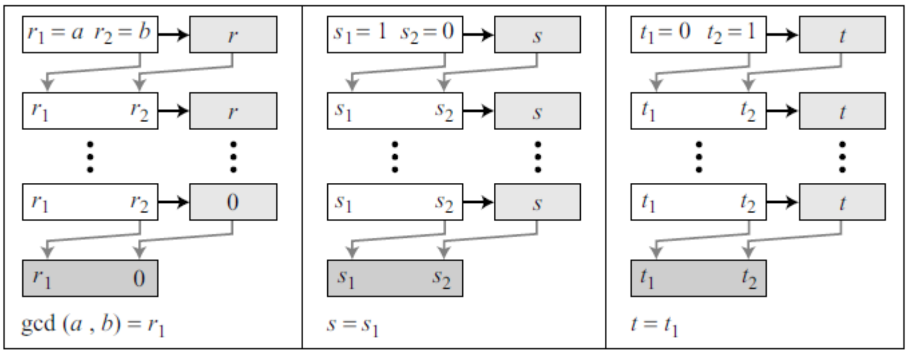
• Find the gcd(25, 60).

q	r_1	r_2	$r = r_1 - q \times r_2$
2	60	25	10
2	25	10	5
2	10	5	0
	$\boxed{5}$	0	

gcd(25, 60)

2. **Write a program to find the Greatest Common Divisor (GCD) of two integers using the Extended Euclidean. Example, say for GCD of 161 and 28, it is essential to find the values of s and t. The details are provided below. (Print GCD, S and t values)**

Extended Euclidean Algorithm:



a. Process

```
r1 ← a;  r2 ← b;
s1 ← 1;  s2 ← 0;          (Initialization)
t1 ← 0;  t2 ← 1;
while (r2 > 0)
{
  q ← r1 / r2;
   r  ← r1 − q × r2;
   r1 ← r2;   r2 ← r;     (Updating r's)
   s  ← s1 − q × s2;
   s1 ← s2;   s2 ← s;     (Updating s's)
   t  ← t1 − q × t2;
   t1 ← t2;   t2 ← t;     (Updating t's)
}
  gcd (a , b) ← r1;   s ← s1;   t ← t1
```

b. Algorithm

•Find the gcd of (161,28) Using Extended Euclidean Algorithm and also find s & t.

q	r_1	r_2	$r = r_1 - r_2 \times q$	s_1	s_2	s	t_1	t_2	t
	161	28		1	0		0	1	
5	161	28	21	1	0	1	0	1	-5
1	28	21	7	0	1	-1	1	-5	6
3	21	7	0	1	-1	4	-5	6	-23
	$\boxed{7}$	0		$\boxed{-1}$	4		$\boxed{6}$	-23	

gcd (161, 28) = 7

$s = s_1 = -1$

$t = t_1 = 6$